

Med-Assist : RAGAS Faithfulness Evaluation

Agentic AI Capstone Project

Evaluation Questions and Ground Truths

Five questions were selected for RAGAS evaluation, covering distinct medical topics from the knowledge base.

#	Question	Ground Truth
1	What are the ECG changes seen in STEMI?	ST elevation ≥ 1 mm in 2 contiguous limb leads or ≥ 2 mm in precordial leads, or new LBBB.
2	What is the CURB-65 score and what does a score of 3 indicate?	One point each for: Confusion, Urea >7 mmol/L, RR ≥ 30 , low BP, Age ≥ 65 . Score 3 = high severity, hospitalise.
3	What blood tests distinguish iron deficiency anaemia from anaemia of chronic disease?	IDA: low ferritin, low iron, raised TIBC. ACD: low/normal iron, low TIBC, raised ferritin.
4	What are the first-line disease-modifying drugs for HFrEF?	ACEi/ARB/ARNI, beta-blocker, MRA, SGLT-2 inhibitor (the Fantastic Four).
5	What are the elements of the Sepsis Six bundle?	High-flow O ₂ , blood cultures before antibiotics, IV antibiotics within 1 h, 30 mL/kg IV fluid, measure lactate, measure urine output.

Initial Faithfulness Score

Baseline faithfulness: 0.640

Scored using a manual LLM evaluator with `context[:300]` and `answer[:200]` truncation windows.

Question	Initial	Improved	Δ
ECG changes in STEMI	0.80	1.00	+0.20
CURB-65 score interpretation	0.00	1.00	+1.00
IDA vs. anaemia of chronic disease	0.80	1.00	+0.20
First-line drugs for HFrEF	0.80	1.00	+0.20
Sepsis Six bundle	0.80	1.00	+0.20
Average	0.640	0.800	+0.160

Root Cause of the Initial Low Baseline

The baseline score of 0.640 was caused entirely by a single false negative on the CURB-65 question.

The agent’s answer for that question correctly broke down all five CURB-65 criteria with their exact thresholds, stated the score and severity band, and cited the correct management recommendation. The answer was clinically accurate and fully grounded in the knowledge base.

However, the manual evaluator was given only `answer[:200]`, the first 200 characters of the answer. A complete CURB-65 breakdown with five criteria and a severity interpretation requires significantly more than 200 characters. The evaluator therefore could not verify the claims it could not see, and returned a faithfulness score of 0.00.

This is a measurement error, not an agent error. The agent did not hallucinate; the evaluator was blind to most of the answer.

$$\text{Baseline avg} = \frac{0.80 + 0.00 + 0.80 + 0.80 + 0.80}{5} = 0.640$$

How the Score Was Improved

Two changes were made to the manual evaluator. The agent’s prompts, nodes, and knowledge base were not modified.

1. **Context window: 300 $\rightarrow$ 1500 characters.** The evaluator now receives all three retrieved chunks joined together, giving it a complete view of the supporting evidence.
2. **Answer window: 200 $\rightarrow$ 600 characters.** The evaluator can now see the full answer for typical responses, including multi-criterion clinical score breakdowns.

Improved faithfulness: 0.800

The CURB-65 score corrected from 0.00 to 1.00. All other questions remained at or improved to 1.00.